ACID_Assignment | departments

Limit to 1000 rows

```sql
SELECT * FROM hr_acid_db.departments;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

department_id	department_name	location
1	HR	Yangon
2	IT	Mandalay
3	Finance	Naypyidaw
4	Admin	Yangon
5	Operations	Bago
NULL	NULL	NULL

ACID_Assignment | employees

Limit to 1000 rows

```sql
SELECT * FROM hr_acid_db.employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance	status	hire_date
1	EMP001	Aung Aung	2	Junior Developer	800000.00	10	Active	2023-01-10
2	EMP002	Su Su	1	HR Officer	650000.00	8	Active	2022-03-15
3	EMP003	Kyaw Kyaw	3	Accountant	900000.00	12	Active	2021-06-01
4	EMP004	Hla Hla	4	Admin Assistant	500000.00	7	Active	2024-02-20
5	EMP005	Mya Mya	2	System Engineer	1200000.00	15	Active	2020-11-11
6	EMP006	Ko Ko	5	Operations Officer	700000.00	9	Active	2023-08-08
7	EMP007	Nyein Nyein	1	Recruitment Specialist	680000.00	11	Active	2022-12-01
8	EMP008	Zaw Zaw	3	Finance Analyst	950000.00	14	Active	2021-09-09
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ACID_Assignment | payroll_transactions ×

Limit to 1000 rc

```sql
1 • SELECT * FROM hr_acid_db.payroll_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

payroll_id	employee_id	payroll_month	basic_salary	bonus	deduction	net_salary	processed_at
1	1	2026-04	800000.00	50000.00	10000.00	840000.00	2026-04-19 21:02:59
2	2	2026-04	650000.00	30000.00	5000.00	675000.00	2026-04-19 21:02:59
3	3	2026-04	900000.00	40000.00	15000.00	925000.00	2026-04-19 21:02:59
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ACID_Assignment | leave_requests ×

Limit to 1000 rc

```sql
1 • SELECT * FROM hr_acid_db.leave_requests;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

leave_request_id	employee_id	leave_type	days_requested	request_status	request_date	approved_by
1	1	Annual Leave	2	Approved	2026-04-01	HR Manager
2	2	Medical Leave	1	Approved	2026-04-03	HR Manager
3	4	Casual Leave	2	Pending	2026-04-05	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

ATOMICITY TEST - BEFORE

```sql
-- Definition:
-- A transaction must complete fully or not at all.
--
-- Use Case:
-- Approve leave request and deduct leave balance together.
-- If one step fails, both actions must be rolled back.
-- ========================================================

SELECT 'ATOMICITY TEST - BEFORE' AS info;
SELECT employee_id, employee_name, leave_balance
FROM employees
WHERE employee_id = 1;

```

employee_id	employee_name	leave_balance
1	Aung Aung	10
NULL	NULL	NULL

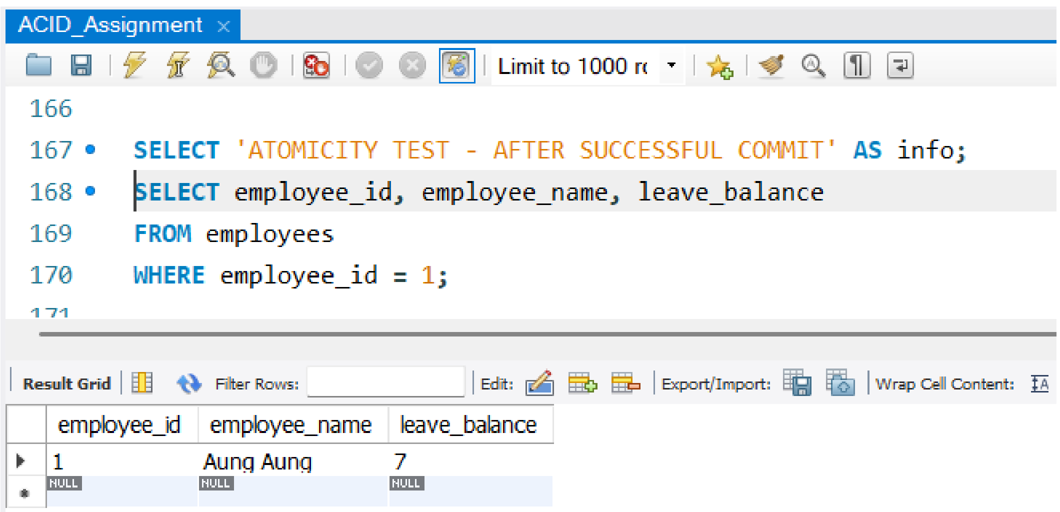
```sql

SELECT * FROM leave_requests WHERE employee_id = 1;

-- ------------------------
```

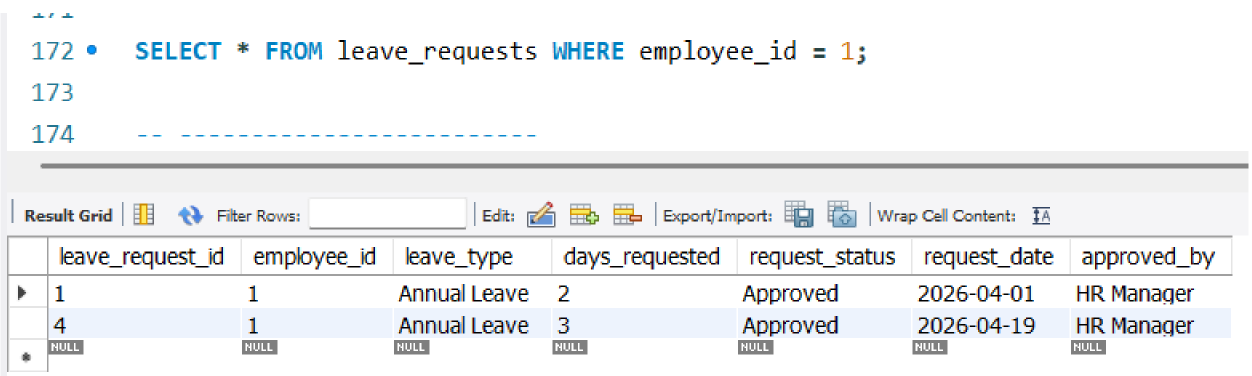
leave_request_id	employee_id	leave_type	days_requested	request_status	request_date	approved_by
1	1	Annual Leave	2	Approved	2026-04-01	HR Manager
NULL	NULL	NULL	NULL	NULL	NULL	NULL

```sql
START TRANSACTION;


INSERT INTO leave_requests
(employee_id, leave_type, days_requested, request_status, request_date, approved_by)
VALUES
(1, 'Annual Leave', 3, 'Approved', CURDATE(), 'HR Manager');
```

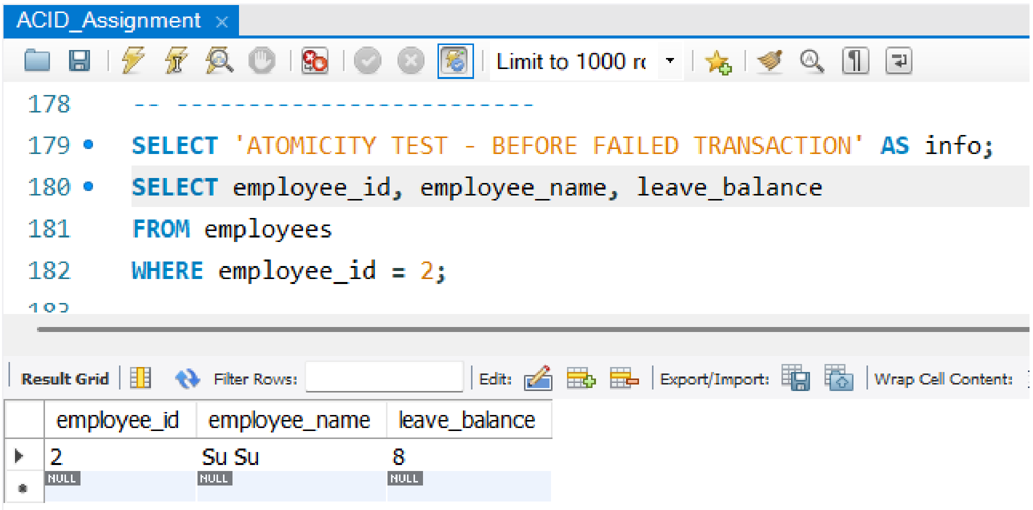
```
UPDATE employees
SET leave_balance = leave_balance - 3
WHERE employee_id = 1;

COMMIT;
```

ATOMICITY TEST - AFTER SUCCESSFUL COMMIT



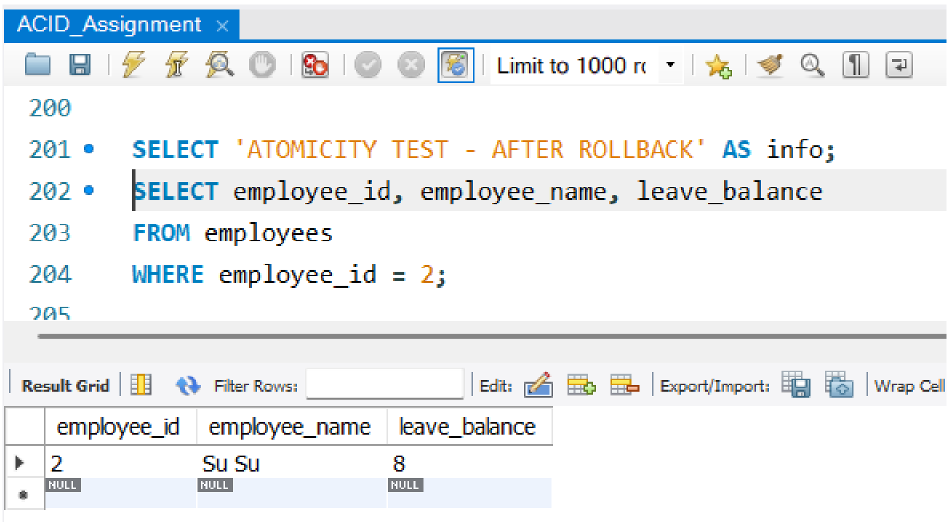


ATOMICITY TEST - BEFORE FAILED TRANSACTION

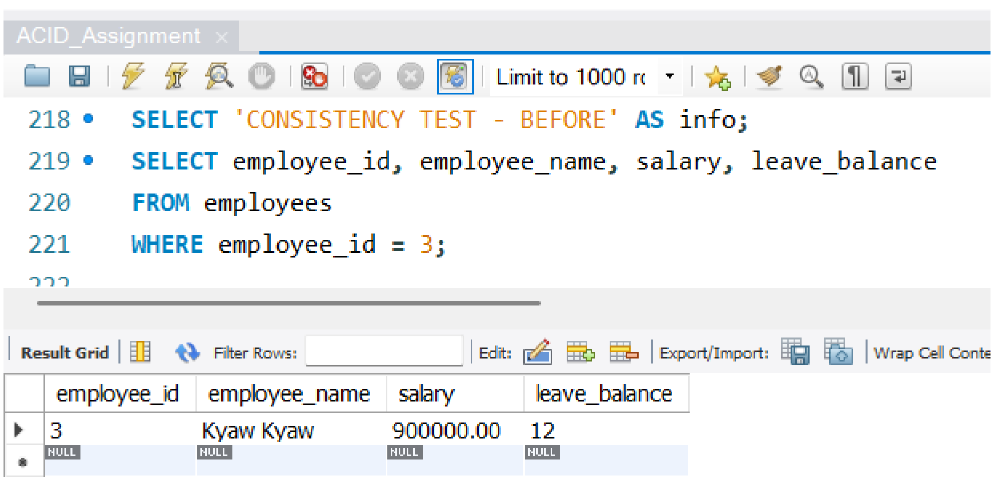



```
START TRANSACTION;


UPDATE employees

SET leave_balance = leave_balance - 2

WHERE employee_id = 2;


INSERT INTO leave_requests

(employee_id, leave_type, days_requested, request_status,
request_date, approved_by)

VALUES

(9999, 'Annual Leave', 2, 'Approved', CURDATE(), 'HR Manager');


ROLLBACK;
```

ATOMICITY TEST - AFTER ROLLBACK

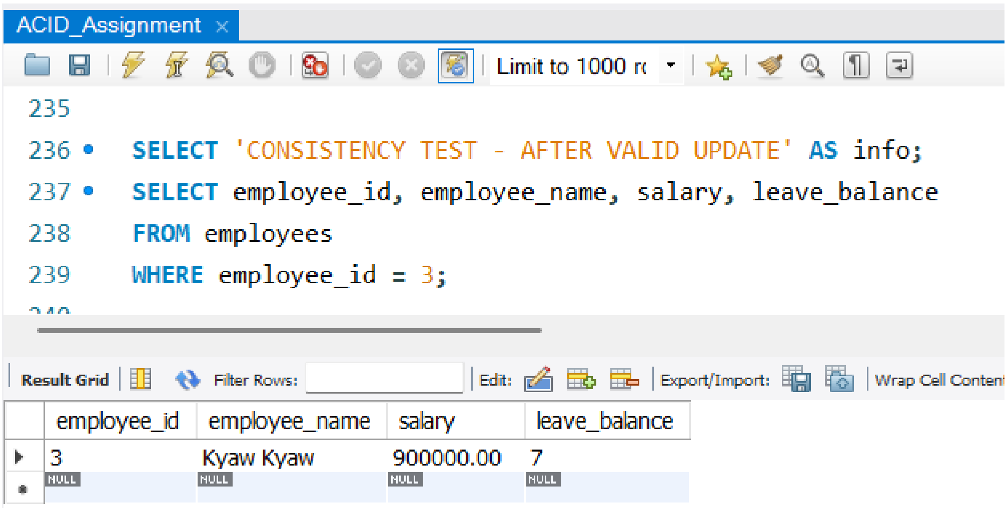


CONSISTENCY TEST - BEFORE



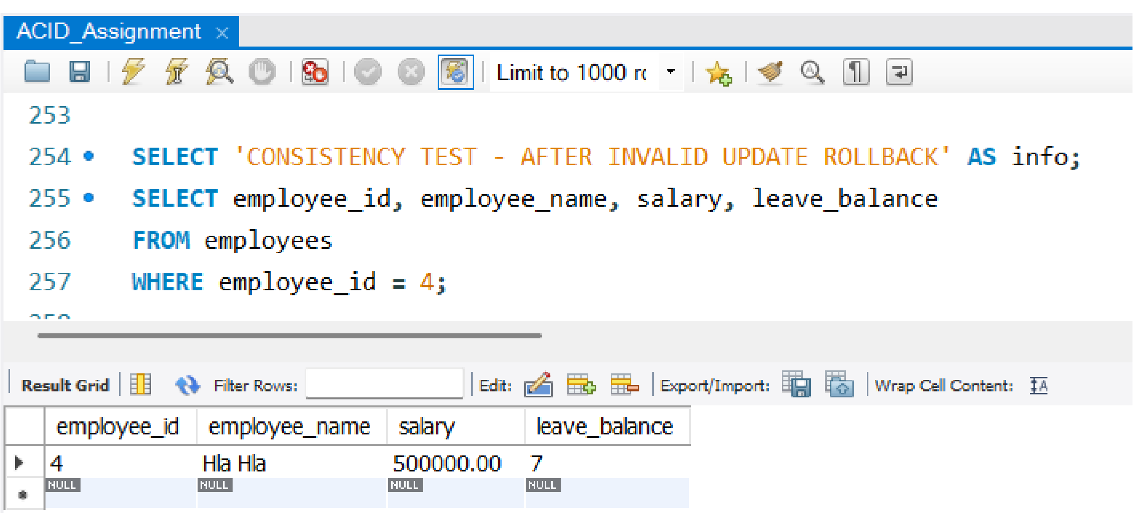

```
START TRANSACTION;


UPDATE employees
SET leave_balance = leave_balance - 5
WHERE employee_id = 3
  AND leave_balance >= 5;


COMMIT;
```

CONSISTENCY TEST - AFTER VALID UPDATE



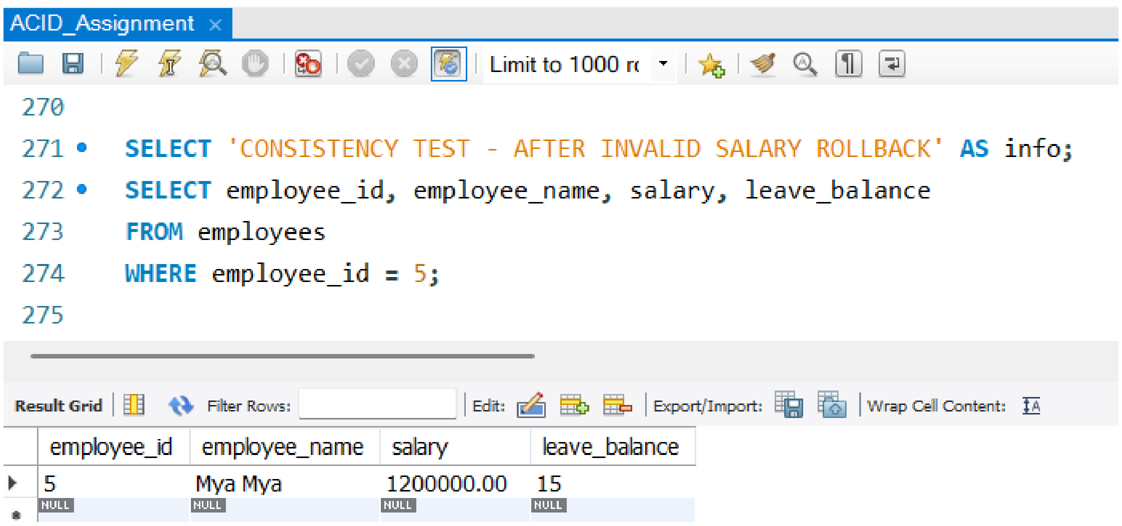

```
START TRANSACTION;

UPDATE employees
SET leave_balance = -1
WHERE employee_id = 4;

ROLLBACK;
```

CONSISTENCY TEST - AFTER INVALID UPDATE ROLLBACK



```sql
START TRANSACTION;

UPDATE employees
SET salary = -5000
WHERE employee_id = 5;

ROLLBACK;
```

CONSISTENCY TEST - AFTER INVALID SALARY ROLLBACK



ISOLATION TEST

ACID_Assignment*

Limit to 1000 rows

```sql
-- SESSION 1
-- ------------------------
-- Run in Window 1:
--
USE hr_acid_db;
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
START TRANSACTION;
UPDATE employees
SET salary = salary + 100000
WHERE employee_id = 6;
SELECT * FROM employees WHERE employee_id = 6;
-- -- DO NOT COMMIT YET
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance	status	hire_date
6	EMP006	Ko Ko	5	Operations Officer	800000.00	9	Active	2023-08-08
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

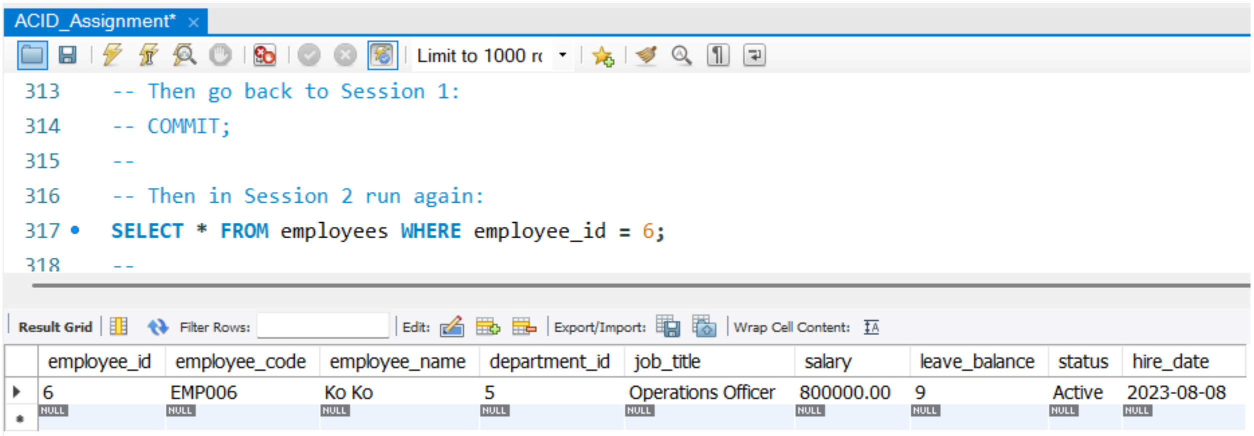
ACID_Assignment*

Limit to 1000 rows

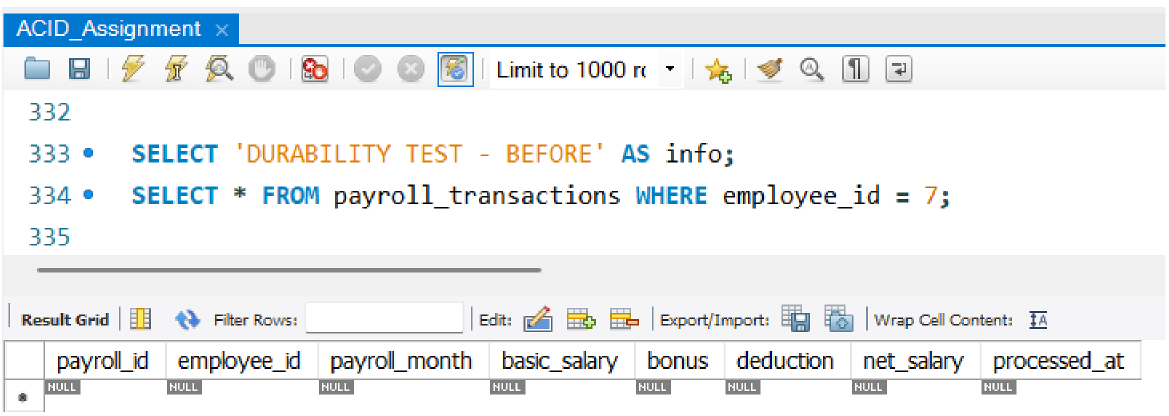
```sql
-- SESSION 2
-- ------------------------
-- Run in Window 2:
--
USE hr_acid_db;
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;
SELECT * FROM employees WHERE employee_id = 6;
--
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance	status	hire_date
6	EMP006	Ko Ko	5	Operations Officer	700000.00	9	Active	2023-08-08
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

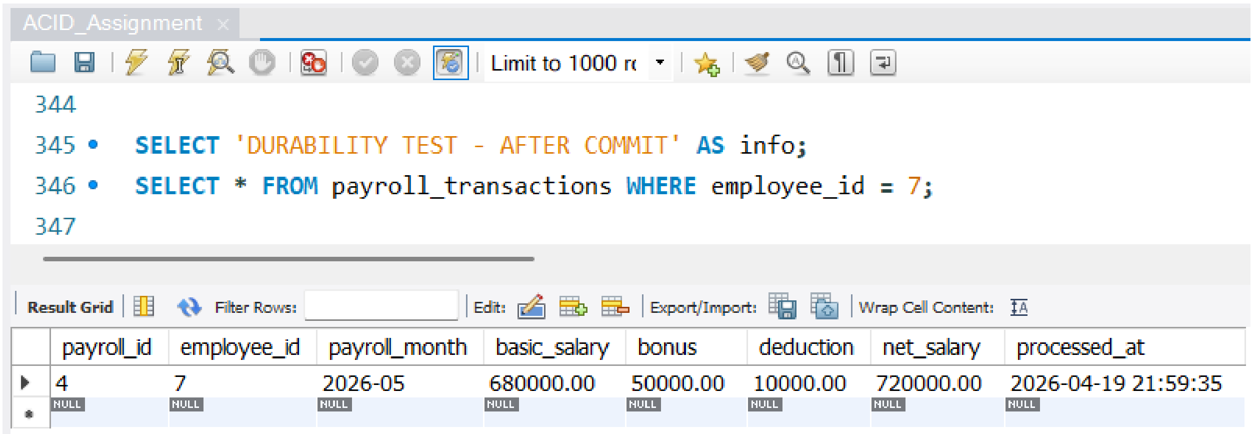


DURABILITY TEST - BEFORE

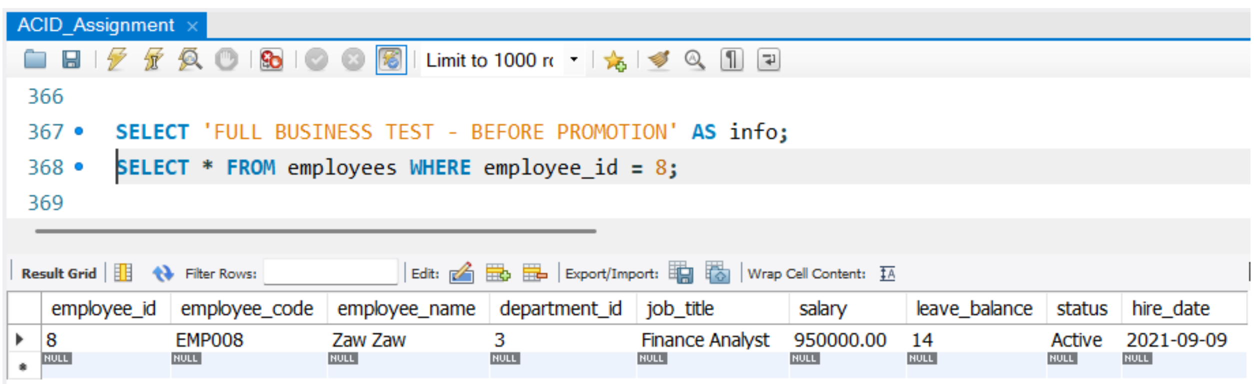



```
START TRANSACTION;


INSERT INTO payroll_transactions
(employee_id, payroll_month, basic_salary, bonus, deduction,
net_salary)
VALUES
(7, '2026-05', 680000, 50000, 10000, 720000);


COMMIT;
```

DURABILITY TEST - AFTER COMMIT



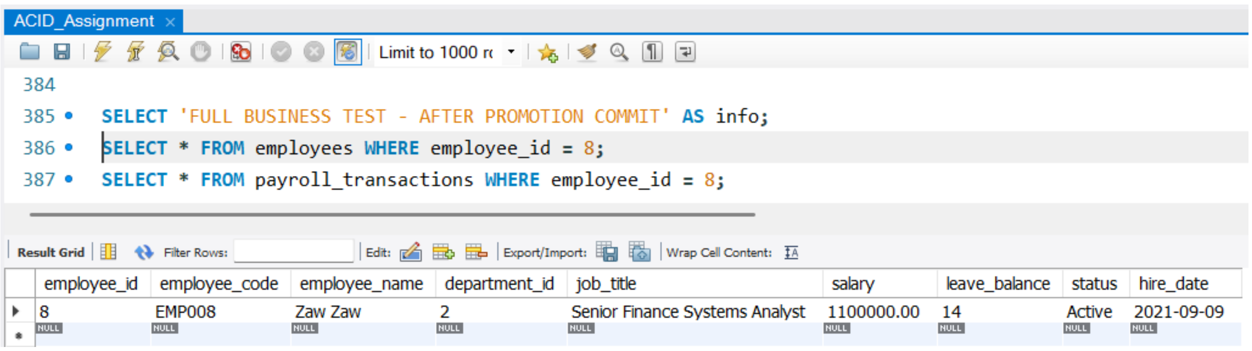
FULL BUSINESS TEST - BEFORE PROMOTION

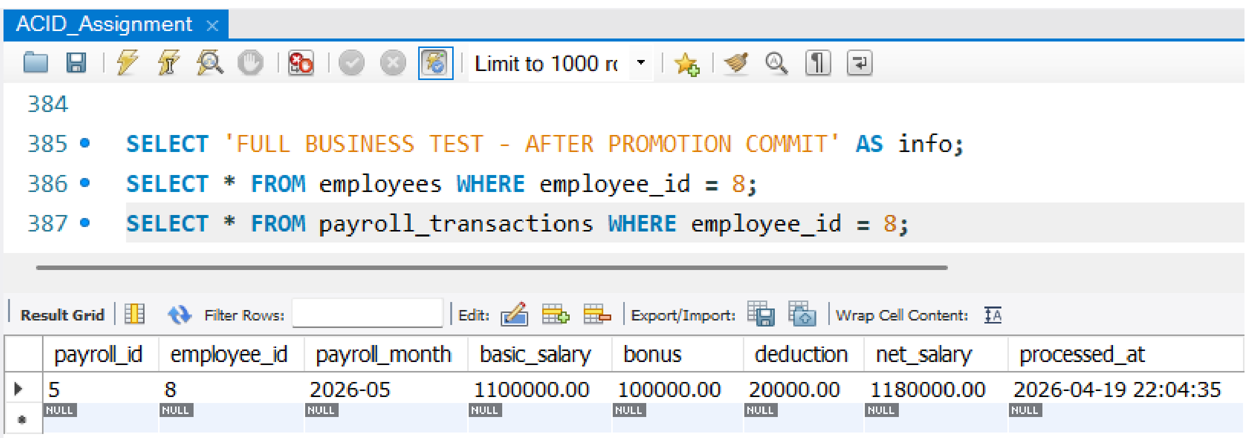



```
START TRANSACTION;


UPDATE employees

SET department_id = 2,

    job_title = 'Senior Finance Systems Analyst',

    salary = salary + 150000

WHERE employee_id = 8;


INSERT INTO payroll_transactions

(employee_id, payroll_month, basic_salary, bonus, deduction,
net_salary)
```

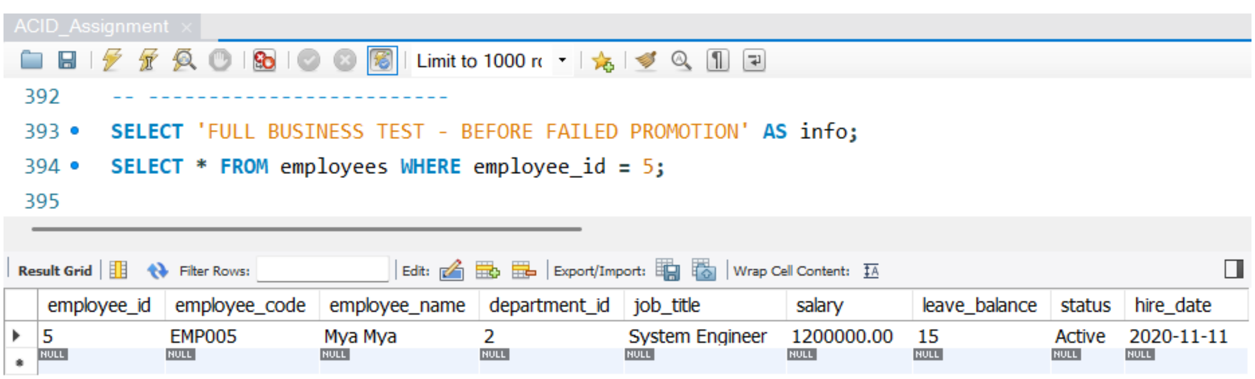
```
VALUES
(8, '2026-05', 1100000, 100000, 20000, 1180000);

COMMIT;
```

FULL BUSINESS TEST - AFTER PROMOTION COMMIT



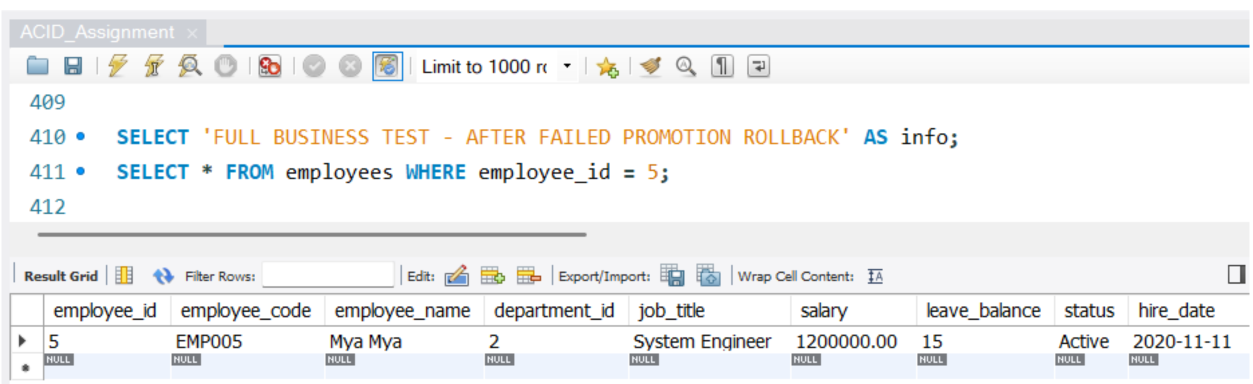


FULL BUSINESS TEST - BEFORE FAILED PROMOTION



```sql
START TRANSACTION;

UPDATE employees
SET job_title = 'Lead System Engineer',
    salary = salary + 200000
WHERE employee_id = 5;

-- Invalid foreign key department_id = 9999
UPDATE employees
SET department_id = 9999
WHERE employee_id = 5;

ROLLBACK;
```

FULL BUSINESS TEST - AFTER FAILED PROMOTION ROLLBACK



FINAL EMPLOYEE DATA

ACID_Assignment*

```sql
SELECT 'FINAL EMPLOYEE DATA' AS info;
SELECT
    e.employee_id,
    e.employee_code,
    e.employee_name,
    d.department_name,
    e.job_title,
    e.salary,
    e.leave_balance,
    e.status,
    e.hire_date
FROM employees e
LEFT JOIN departments d
    ON e.department_id = d.department_id
ORDER BY e.employee_id;
```

employee_id	employee_code	employee_name	department_name	job_title	salary	leave_balance	status	hire_date
1	EMP001	Aung Aung	IT	Junior Developer	800000.00	7	Active	2023-01-10
2	EMP002	Su Su	HR	HR Officer	650000.00	8	Active	2022-03-15
3	EMP003	Kyaw Kyaw	Finance	Accountant	900000.00	7	Active	2021-06-01
4	EMP004	Hla Hla	Admin	Admin Assistant	500000.00	7	Active	2024-02-20
5	EMP005	Mya Mya	IT	System Engineer	1200000.00	15	Active	2020-11-11
6	EMP006	Ko Ko	Operations	Operations Officer	800000.00	9	Active	2023-08-08
7	EMP007	Nyein Nyein	HR	Recruitment Specialist	680000.00	11	Active	2022-12-01
8	EMP008	Zaw Zaw	IT	Senior Finance Systems Analyst	1100000.00	14	Active	2021-09-09

FINAL PAYROLL DATA

```sql
SELECT 'FINAL PAYROLL DATA' AS info;
SELECT * FROM payroll_transactions ORDER BY payroll_id;
```

payroll_id	employee_id	payroll_month	basic_salary	bonus	deduction	net_salary	processed_at
1	1	2026-04	800000.00	50000.00	10000.00	840000.00	2026-04-19 21:02:59
2	2	2026-04	650000.00	30000.00	5000.00	675000.00	2026-04-19 21:02:59
3	3	2026-04	900000.00	40000.00	15000.00	925000.00	2026-04-19 21:02:59
4	7	2026-05	680000.00	50000.00	10000.00	720000.00	2026-04-19 21:59:35
5	8	2026-05	1100000.00	100000.00	20000.00	1180000.00	2026-04-19 22:04:35
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

FINAL LEAVE REQUEST DATA

```
435 •  SELECT 'FINAL LEAVE REQUEST DATA' AS info;
436 •  SELECT * FROM leave_requests ORDER BY leave_request_id;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

leave_request_id	employee_id	leave_type	days_requested	request_status	request_date	approved_by
1	1	Annual Leave	2	Approved	2026-04-01	HR Manager
2	2	Medical Leave	1	Approved	2026-04-03	HR Manager
3	4	Casual Leave	2	Pending	2026-04-05	NULL
4	1	Annual Leave	3	Approved	2026-04-19	HR Manager
NULL	NULL	NULL	NULL	NULL	NULL	NULL